Light

GCSE Computer Science — Character Sets (Set 1)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is a character set?

- ○ **A.** A collection of all possible binary numbers
- ○ **B.** A defined list of characters that a computer recognises, where each character is assigned a unique binary code
- ○ **C.** A set of fonts used by word processors
- ○ **D.** A group of keyboard shortcuts

Q2 (1 mark)

Why must characters be converted to binary for computer storage?

- ○ **A.** Because computers can only process and store data as binary (0s and 1s)
- ○ **B.** Because binary is easier for humans to read
- ○ **C.** Because binary uses less electricity than other number systems
- ○ **D.** Because keyboards only produce binary output

Q3 (1 mark)

What does ASCII stand for?

- ○ **A.** American Standard Code for Information Interchange
- ○ **B.** Advanced System Code for Internet Integration
- ○ **C.** Automatic String Conversion for Input Interface
- ○ **D.** American Symbolic Characters for Industrial Interchange

Q4 (1 mark)

In the OCR examination, ASCII characters are represented using how many bits?

- ○ **A.** 4 bits
- ○ **B.** 6 bits
- ○ **C.** 7 bits
- ○ **D.** 8 bits

Q5 (1 mark)

What is Unicode?

- ○ **A.** A programming language for creating websites
- ○ **B.** A character set that can represent characters from virtually all writing systems worldwide
- ○ **C.** A type of computer virus
- ○ **D.** A method of compressing text files

Q6 (1 mark)

In ASCII, if the character 'A' has the code 65, what code does 'B' have?

- ○ **A.** 64
- ○ **B.** 66
- ○ **C.** 67
- ○ **D.** 75

Q7 (1 mark)

Using 7 bits per character, how many different characters can be represented?

- ○ **A.** 7 characters
- ○ **B.** 49 characters
- ○ **C.** 128 characters

- ○ **D.** 256 characters

Q8 (1 mark)

Which types of characters are included in the ASCII character set?

- ○ **A.** Only uppercase letters
- ○ **B.** Only numbers 0-9
- ○ **C.** Letters (uppercase and lowercase), numbers, punctuation, and control characters
- ○ **D.** Only Chinese and Japanese characters

Q9 (1 mark)

Why does Unicode typically use more bits per character than ASCII?

- ○ **A.** To represent a much larger number of characters from different languages and scripts
- ○ **B.** To make files load faster
- ○ **C.** To reduce file sizes
- ○ **D.** To improve internet connection speeds

Q10 (1 mark)

In ASCII, lowercase 'a' has code 97. What is the code for lowercase 'd'?

- ○ **A.** 98
- ○ **B.** 99
- ○ **C.** 100
- ○ **D.** 101

Q11 (1 mark)

What is a key advantage of ASCII over Unicode?

- ○ **A.** ASCII can represent more characters
- ○ **B.** ASCII supports all world languages
- ○ **C.** ASCII uses less storage space per character
- ○ **D.** ASCII was developed more recently

Q12 (1 mark)

A character set uses 5 bits per character. How many unique characters can it represent?

- ○ **A.** 5 characters
- ○ **B.** 10 characters
- ○ **C.** 25 characters
- ○ **D.** 32 characters

Q13 (2 marks)

A text file contains 500 characters stored using ASCII (8 bits per character). Calculate the file size in bytes.

- ○ **A.** 62.5 bytes
- ○ **B.** 4,000 bytes
- ○ **C.** 500 bits
- ○ **D.** 500 bytes

Q14 (2 marks)

A document contains 200 characters stored using Unicode (16 bits per character). Calculate the file size in bytes.

- ○ **A.** 400 bytes
- ○ **B.** 200 bytes
- ○ **C.** 1,600 bytes
- ○ **D.** 3,200 bytes

Q15 (2 marks)

In ASCII, 'M' has the code 77. A programmer needs to find the code for 'R'. Explain how to calculate this.

- ○ **A.** Add 10 to 77 because R is the 10th letter after M
- ○ **B.** Subtract 5 from 77 because R comes before M
- ○ **C.** Multiply 77 by 2 because R is twice as far in the alphabet

○ **D.** Add 5 to 77 because R is 5 letters after M in the alphabet, giving 82

Q16 (2 marks)

A website needs to display text in English, Chinese, and Arabic. Explain why Unicode is necessary instead of ASCII.

○ **A.** Unicode loads faster than ASCII on websites

○ **B.** ASCII cannot represent Chinese or Arabic characters because its 128-256 characters only cover English letters, numbers, and basic symbols

○ **C.** Unicode is older and more established than ASCII

○ **D.** ASCII files are too large for web transmission

Q17 (2 marks)

A company creates a custom character set for a device with limited memory. They need exactly 64 characters. What is the minimum number of bits required per character?

○ **A.** 5 bits

○ **B.** 7 bits

○ **C.** 8 bits

○ **D.** 6 bits

Q18 (2 marks)

Two identical text documents contain 1,000 characters. Document A uses ASCII (8 bits), Document B uses Unicode (16 bits). How much larger is Document B?

- ○ **A.** Document B is the same size
- ○ **B.** Document B is 500 bytes larger
- ○ **C.** Document B is 1,000 bytes larger
- ○ **D.** Document B is 2,000 bytes larger

Q19 (2 marks)

A text message uses the character code 72. If the message contains the sequence 72, 69, 76, 76, 79, what word does it spell?

- ○ **A.** WORLD
- ○ **B.** HELP
- ○ **C.** HERO
- ○ **D.** HELLO

Q20 (2 marks)

UTF-8 is a variable-length Unicode encoding. What does 'variable-length' mean in this context?

- ○ **A.** Files can be any size
- ○ **B.** Different characters may use different numbers of bytes (1-4 bytes depending on the character)
- ○ **C.** The encoding changes depending on the day

- D. Characters can have multiple meanings

Q21 (2 marks)

A messaging app needs to support 50,000 different emoji characters. What is the minimum number of bits required per emoji?

- A. 15 bits
- B. 17 bits
- C. 50 bits
- D. 16 bits

Q22 (2 marks)

In ASCII, 'A' = 65 and 'a' = 97. What is the decimal difference between any uppercase letter and its lowercase equivalent?

- A. 16
- B. 32
- C. 26
- D. 64

Q23 (2 marks)

A legacy database system only supports ASCII. A company wants to store customer names in Japanese. What problem will occur?

- A. The database will run slower

○ **B.** Japanese characters will display as incorrect symbols, question marks, or be rejected entirely because ASCII cannot encode them

○ **C.** The database will automatically convert to Unicode

○ **D.** Japanese names will be stored correctly but in a different font

Q24 (3 marks)

A novel contains 400,000 characters. Compare the file sizes when stored as:

- ASCII (8 bits per character)
- Unicode UTF-16 (16 bits per character)

Analyse the storage implications.

○ **A.** ASCII: 400 KB, UTF-16: 800 KB; UTF-16 requires exactly double the storage, making ASCII preferable for English-only text where storage is limited

○ **B.** ASCII: 400 KB, UTF-16: 400 KB; both use the same storage because they encode the same text

○ **C.** ASCII: 800 KB, UTF-16: 400 KB; UTF-16 is more efficient due to modern compression

○ **D.** ASCII: 3.2 MB, UTF-16: 6.4 MB; both formats are impractically large for text storage

Q25 (3 marks)

A developer receives a text file that displays 'Ã©' instead of 'é'. Analyse what has caused this problem and how character encoding relates to it.

○ **A.** The file is corrupted and cannot be recovered

○ **B.** The file was saved in one encoding (likely UTF-8) but opened using a different encoding (likely ASCII or Latin-1); the multi-byte UTF-8

sequence for 'é' is being interpreted as separate ASCII characters

○ **C.** The keyboard that created the file was broken

○ **D.** The character 'é' does not exist in any character set

Q26 (3 marks)

A sorting algorithm arranges customer names alphabetically using ASCII codes. The names 'Smith', 'JONES', and 'adams' are sorted.
Analyse why the result might be unexpected and explain the underlying principle.

○ **A.** The result is JONES, Smith, adams because all uppercase letters (65-90) have lower codes than all lowercase letters (97-122), so uppercase sorts before lowercase regardless of alphabetical order

○ **B.** The result is adams, JONES, Smith because lowercase always sorts first

○ **C.** The result is correctly alphabetical: adams, JONES, Smith

○ **D.** ASCII cannot be used for alphabetical sorting

Q27 (3 marks)

A global social media platform must support posts in all languages including emoji. The platform stores 1 billion posts per day.
Analyse the character encoding choice and its implications.

○ **A.** Use ASCII to minimise storage costs; users can type phonetically

○ **B.** Use a fixed 32-bit Unicode encoding for every character to ensure universal compatibility, accepting the storage overhead

○ **C.** Use UTF-8 variable encoding: ASCII characters use 1 byte while complex characters use 2-4 bytes, balancing storage efficiency with

universal language support; most English text remains compact while full Unicode range is available

○ **D.** Create a custom character set specific to the platform's needs

Q28 (3 marks)

A telecommunications company designs a messaging protocol. They must support 200 characters and want to maximise the number of messages that fit in a fixed-size transmission.
Analyse the optimal number of bits per character.

○ **A.** Use 7 bits (128 characters) and exclude some characters to fit more messages

○ **B.** Use 8 bits (256 characters) because it's a standard byte size, even though it provides more characters than needed

○ **C.** Use exactly 8 bits because powers of 2 are required for character encoding

○ **D.** Use 9 bits (512 characters) to allow for future expansion

Q29 (3 marks)

A web application validates user input by checking for dangerous characters like '<' and '>'. An attacker submits visually identical characters from different Unicode ranges.
Analyse this security vulnerability.

○ **A.** This is not a real security concern because all similar-looking characters have the same code

○ **B.** Unicode includes multiple characters that appear identical but have different codes; the application may block ASCII '<' (code 60) but accept Unicode 'fullwidth less-than sign' (code 65308) which renders identically, potentially enabling injection attacks

○ **C.** Unicode characters cannot be used in web applications

○ **D.** The browser automatically converts all Unicode to ASCII for security

Q30 (3 marks)

Examine why ASCII was sufficient for early computing but became inadequate as technology globalised.
Analyse the historical and technical factors.

○ **A.** ASCII became inadequate only because computers became faster and needed more characters to process

○ **B.** ASCII was never adequate; it was a temporary solution that was always planned to be replaced

○ **C.** ASCII was designed in 1960s America primarily for English text and telegraph communication; as computing spread globally, the 128-character limit could not accommodate thousands of Chinese characters, Arabic scripts, or other writing systems, necessitating Unicode's development in the 1990s

○ **D.** ASCII is still adequate; Unicode is only used for emoji

Total: **55** marks • Time: **30 min**

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